

VAVA'U, Tonga

WMO: 917790

Lat: 18.583S

Lon: 173.967W

Elev: 70

StdP: 100.49

Time Zone: 13.00 (E13)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.0	17.5	13.2	9.5	20.6	14.2	10.2	20.8	9.1	24.7	8.3	24.0	0.1	140	0.323

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.9	30.2	26.3	30.0	26.1	29.7	25.9	27.1	29.3	26.7	28.9	26.4	28.6	4.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.8	28.6	26.1	21.6	28.5	25.8	21.3	28.3	86.0	29.9	84.5	29.1	83.0	28.5	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.6	7.6	7.0	14.0	31.0	1.2	0.6	13.1	31.4	12.5	31.8	11.8	32.1	11.0	32.6
			13.1	27.8	0.8	0.4	12.6	28.1	12.1	28.3	11.7	28.5	11.1	28.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	26.5	26.8	26.5	25.8	24.7	23.9	23.2	23.1	23.7	24.3	25.0	25.9
	DBStd	1.90	1.09	1.10	1.05	1.27	1.47	1.46	1.66	1.58	1.63	1.56	1.53	1.32
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5455	511	471	513	474	456	417	408	408	411	444	450	493
	CDD18.3	2413	252	238	254	224	197	167	149	149	161	186	200	235
	CDH23.3	18117	2429	2364	2461	1950	1325	834	613	609	808	1103	1525	2096
	CDH26.7	3003	551	590	574	350	116	48	22	27	53	91	197	383
(14) Wind	WSAvg	3.4	3.4	3.6	2.8	3.2	3.4	3.4	3.3	3.4	3.5	3.6	3.5	3.6
(15) Precipitation	PrecAvg	2170	287	224	269	220	151	136	101	95	141	171	146	202
	PrecMax	3424	618	487	619	585	372	529	234	335	423	426	478	530
	PrecMin	1066	46	64	63	47	28	0	15	9	1	0	5	34
	PrecStd	514	185	99	132	136	94	134	60	70	113	113	107	127
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.9	31.2	30.7	30.2	29.1	28.8	28.1	28.1	28.8	29.0	29.5	30.2
		MCWB	26.7	26.9	26.7	26.1	25.4	25.4	24.7	24.3	25.1	24.9	25.4	26.0
	2%	DB	30.1	30.5	30.1	29.7	28.3	27.9	27.2	27.3	27.9	28.1	29.0	29.7
		MCWB	26.1	26.6	26.3	25.7	24.9	24.8	24.1	23.8	24.1	24.3	25.2	25.7
	5%	DB	29.8	30.0	29.8	29.1	27.9	27.1	26.7	26.9	27.1	27.8	28.4	29.1
		MCWB	25.9	26.3	26.1	25.3	24.6	24.0	23.6	23.5	23.5	24.1	24.7	25.3
	10%	DB	29.1	29.3	29.1	28.5	27.2	26.3	26.0	26.0	26.5	27.1	27.9	28.8
		MCWB	25.6	25.9	25.7	25.0	24.0	23.5	23.0	22.9	23.1	23.7	24.4	25.0
	0.4%	WB	27.2	27.5	27.5	26.8	26.2	26.1	25.6	25.4	25.8	25.7	26.5	27.0
		MCDB	29.4	29.8	29.5	28.9	28.0	27.8	27.1	27.0	28.0	27.6	28.4	28.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.8	27.1	26.9	26.2	25.7	25.4	25.0	24.7	24.9	25.2	25.9	26.2
		MCDB	29.0	29.5	29.1	28.5	27.5	27.1	26.6	26.5	26.8	27.2	28.0	28.4
	5%	WB	26.3	26.7	26.5	25.9	25.3	24.9	24.4	24.2	24.3	24.7	25.5	25.9
		MCDB	28.5	29.0	28.7	28.3	27.1	26.5	26.0	26.0	26.3	26.8	27.6	28.2
	10%	WB	26.0	26.3	26.2	25.6	24.9	24.3	23.8	23.7	23.8	24.3	25.1	25.6
		MCDB	28.2	28.5	28.4	27.8	26.7	25.9	25.4	25.5	25.7	26.3	27.2	27.8
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	4.9	5.3	5.4	5.1	4.7	5.1	5.2	5.3	5.0	5.3	5.2
		MCDBR	6.1	5.7	5.7	6.2	5.5	5.6	5.8	6.0	5.9	5.6	5.8	5.5
		MCWBR	3.2	3.0	3.0	3.4	3.2	3.5	3.7	3.6	3.5	3.2	3.3	2.9
	5% WB	MCDBR	5.1	4.9	5.1	5.4	4.9	4.7	4.7	5.0	4.9	4.9	5.1	5.1
		MCWBR	3.0	2.7	3.0	3.1	3.2	3.3	3.4	3.4	3.3	3.1	3.3	3.0
(40) Clear-Sky Solar Irradiance	taub		0.384	0.379	0.373	0.394	0.382	0.384	0.366	0.375	0.397	0.382	0.380	0.386
	taud		2.626	2.644	2.679	2.594	2.610	2.580	2.625	2.599	2.537	2.582	2.610	2.600
	Ebn at Noon		960	954	938	880	855	833	862	886	900	940	957	958
	Edh at Noon		102	99	92	94	87	86	84	92	105	104	103	105
(44) All-Sky Solar Radiation	RadAvg		6.29	5.98	5.72	4.80	4.20	3.80	4.14	4.69	5.32	5.92	6.22	6.40
	RadStd		0.66	0.63	0.39	0.49	0.27	0.33	0.23	0.30	0.36	0.42	0.48	0.46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page